

CY 401 : Introduction to Quantum Chemistry & Molecular Spectroscopy

Course Code : CY 401

Course Name : Introduction to Quantum Chemistry & Molecular Spectroscopy

L-T-P-C : 3-0-0-3

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Module I:** Introduction to Schrödinger equation, Bohr's atom, De Broglie's Wave, wave-particle duality, Light-atom/molecule interaction, Introduction to optical spectroscopy, time- independent Schrodinger equation, Particle in a box, Quantum Mechanics of Hydrogen Atom. (20 Hours)
- **Module II:** Heisenberg's Uncertainty Relation, Operators, Commutators, Eigenvalues and Eigenvectors, absorption and emission spectra, Boltzmann Energy distribution, Principle of equipartition of energy Einstein's Semiclassical model, Born Oppenheimer Approximation, Beer-Lambert Law. (14 Hours)
- **Module III:** Diatomic Vibrational Spectra: Harmonic Model, Morse Oscillator Model, Molecular Vibrations in Polyatomic Molecules, Diatomic rotational spectra, rotation of polyatomic molecules, electronic absorption, and emission spectra. (8 Hours)

Textbooks:

1. Peter Atkins, Julio de Paula, James Keeler, **Physical Chemistry**, Oxford University Press, 2018.

References:

1. Thomas Engel, **Quantum Chemistry & Spectroscopy**, 3rd Edition, Pearson, 2015.
2. Donald A. McQuarrie & John D. Simons, **Physical Chemistry: A molecular approach**, Indian Reprint, Viva books, 2019.